

Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	The user interface was developed using HTML and CSS because they are simple, lightweight, and suitable for creating responsive web pages where users can enter applicant details and view prediction results.
2	Backend / Server-Side	Python, Flask	Python was chosen because of its excellent machine learning support. Flask was used as the backend framework to connect the web interface with the trained machine learning model and process prediction requests efficiently.

3	Database / Data Storage	CSV Datasets (application_record.csv, credit_record.csv, final_credit_dataset.csv)	Instead of using a database, CSV files were used to store the training data. This approach is simple and suitable for an academic machine learning project.
4	Cloud / Hosting / Deployment	Render	Render was selected because it provides free and easy deployment for Flask applications directly from GitHub, making the project accessible online.
5	Version Control & CI/CD	Git and GitHub	Git was used to track code changes, while GitHub was used to store the project repository, collaborate with team members, and manage version control.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, XGBoost, Joblib, Imbalanced-learn (SMOTE), Matplotlib	Scikit-learn was used to build machine learning models. Pandas and NumPy were used for data preprocessing. XGBoost and Random Forest were used for model comparison. SMOTE handled class imbalance. Joblib saved the trained model, and Matplotlib was used for model evaluation and visualization